

Local Liveability Index

A liveability score for *any* location in India.

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Problem:

- Currently liveability is calculated only for the top 10 cities in India and they are ranked.
- There is no way to compare liveability of different areas ***within*** cities.

What is Liveability?

- Liveability is a measure of the quality of life in a place, encompassing factors like social, economic, and environmental conditions that make a location favorable for living.
- It includes aspects such as safety, housing, infrastructure, healthcare, education, and the overall cultural and social atmosphere.

Our Solution

- Latitude and Longitude is taken as input.
- Data is collected from various sources about the liveability of the specified location.
- The location is scored out of 100 based on the data collected.
- The results are plotted on a map.
- Users can also specifically select the parameters they are interested in.

Parameters Used:

Environment & Sustainability

- Air Quality (AQI)
- Water Quality (WQI)
- Parks
- Playgrounds

Infrastructure & Services

- Hospitals
- Schools
- Supermarkets
- Stores
- Libraries

Mobility & Accessibility

- City Congestion
- Local Congestion

Economy & Affordability

- Property Prices
- HDI Rank

Civic Engagement

- Voter Turnout

Methodology:

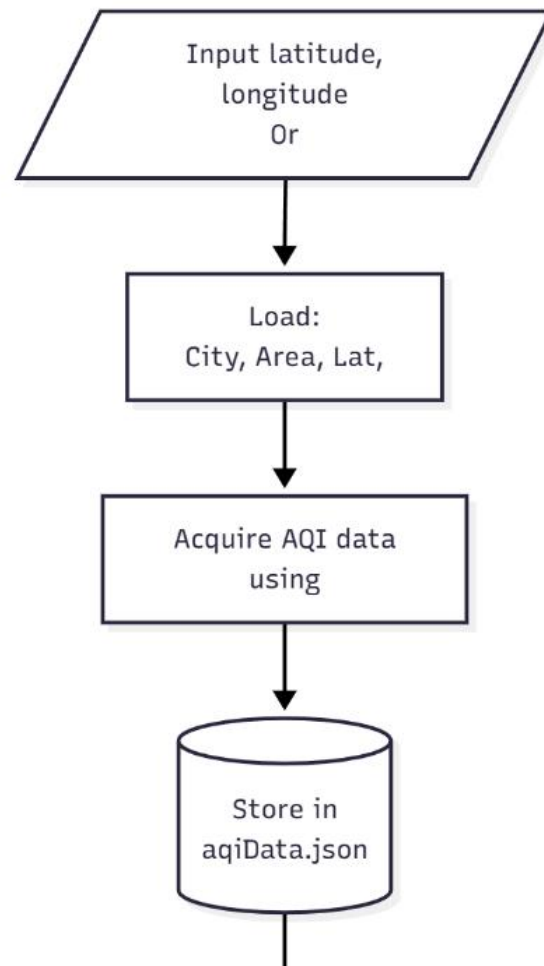
Score Calculation:

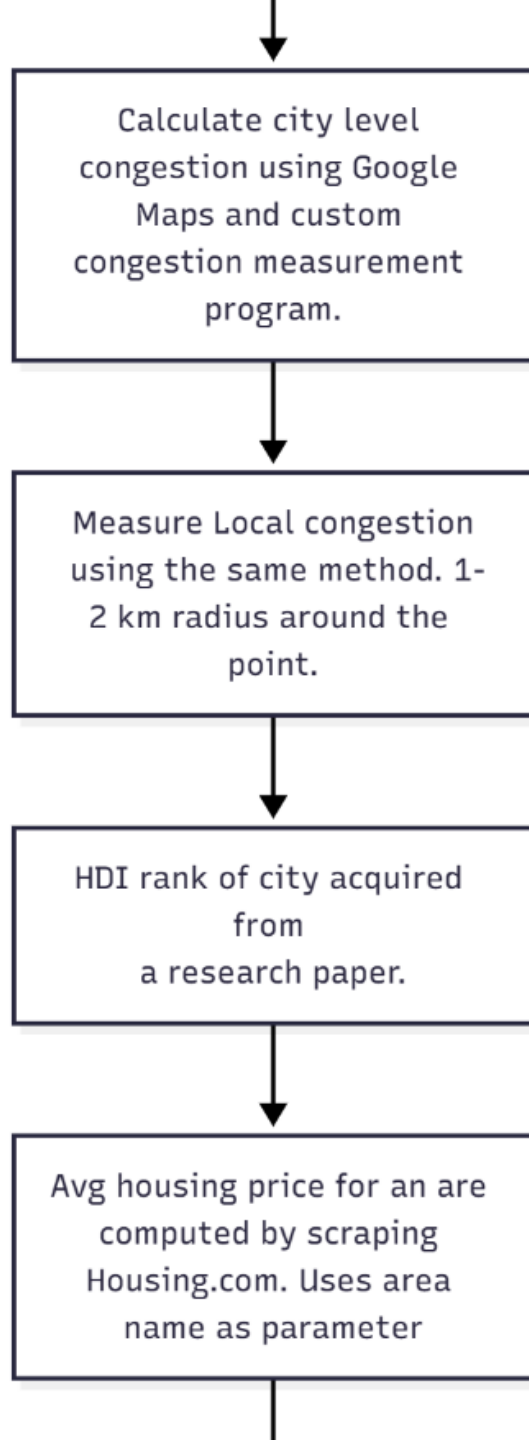
- Score is calculated via weighted sum.
- Cross verified with Principal Component Analysis.

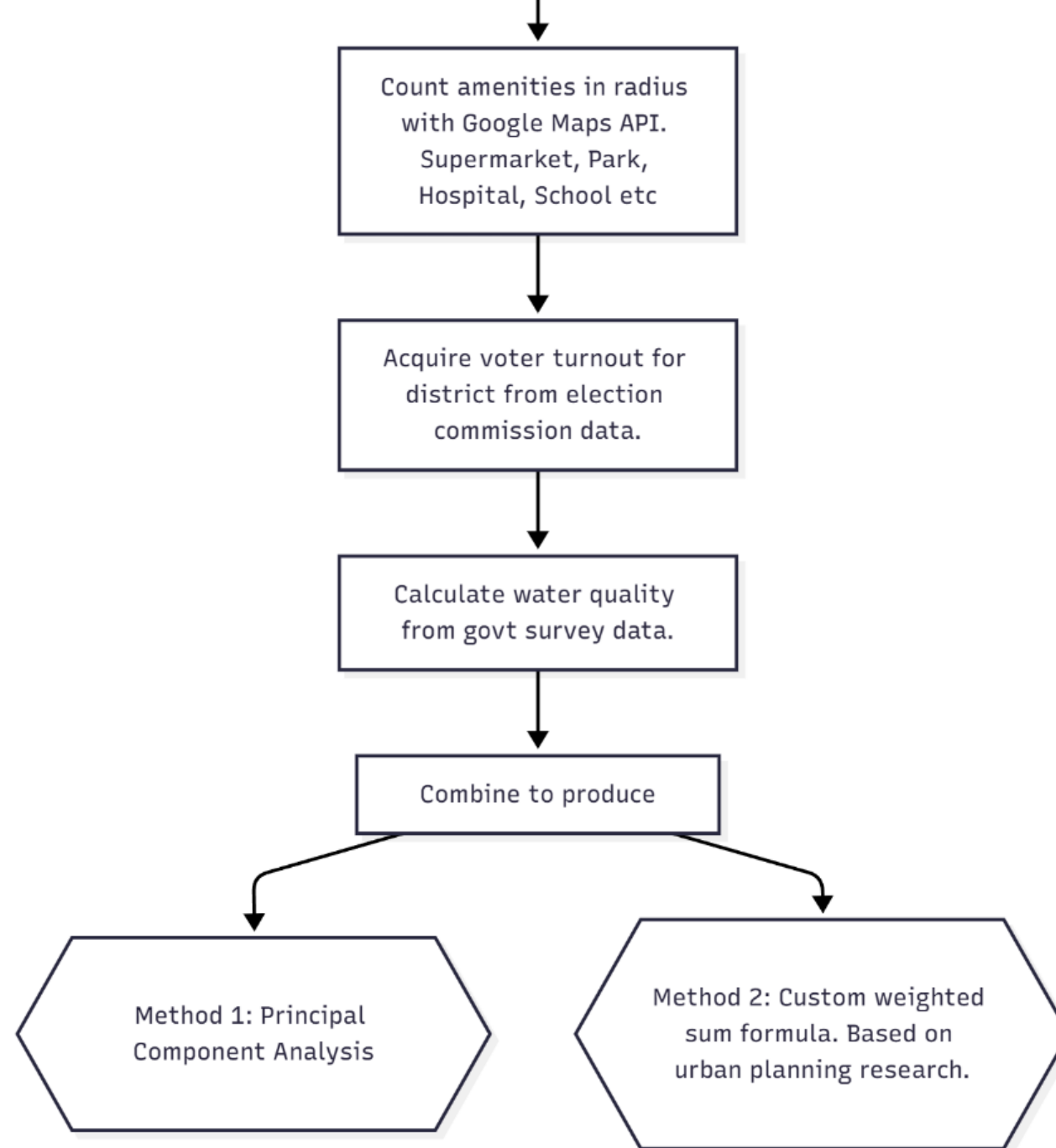
Data Sources:

- Google Maps, Housing.com, WAQI.com, data.gov.in etc.

Process Flowchart

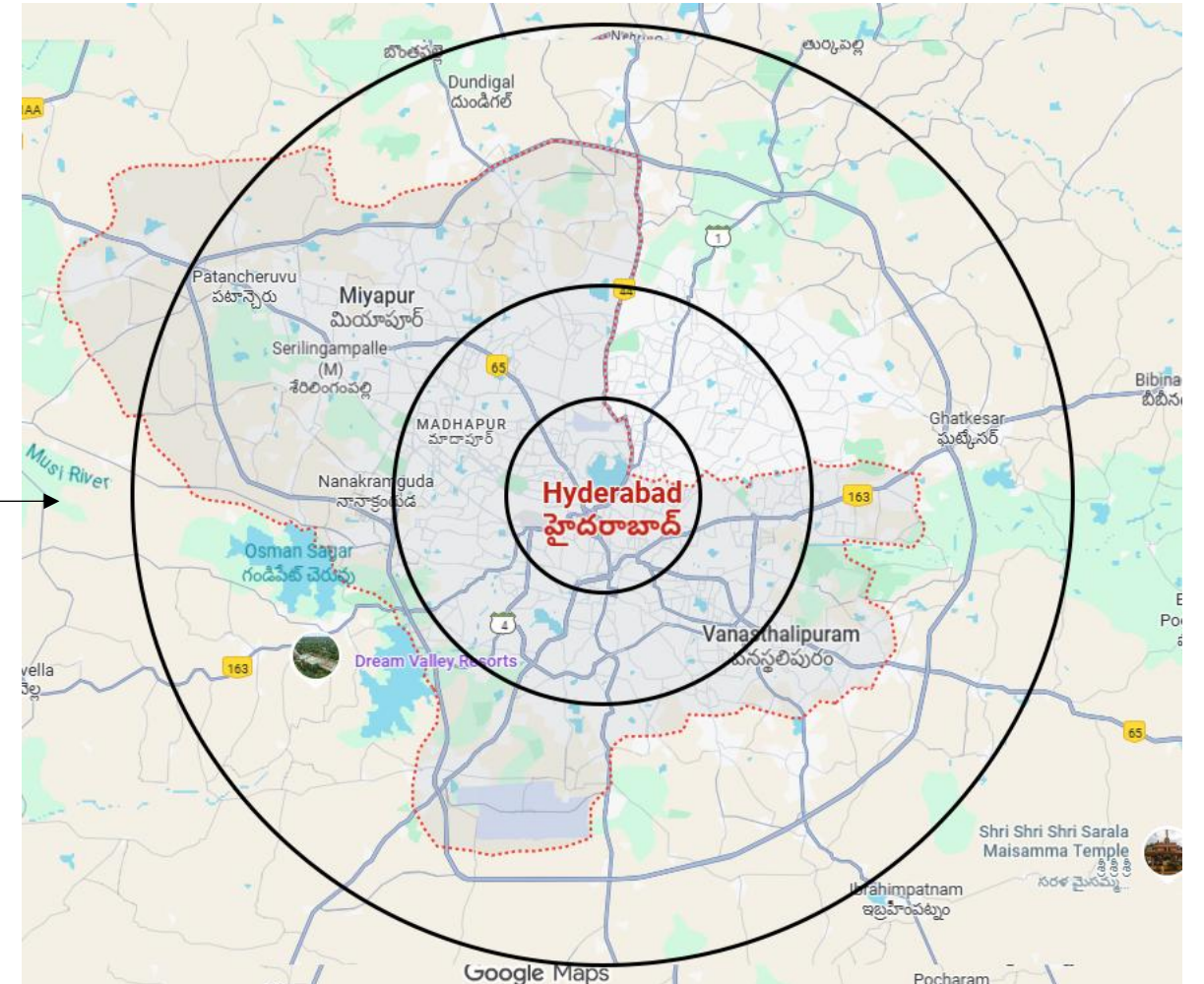
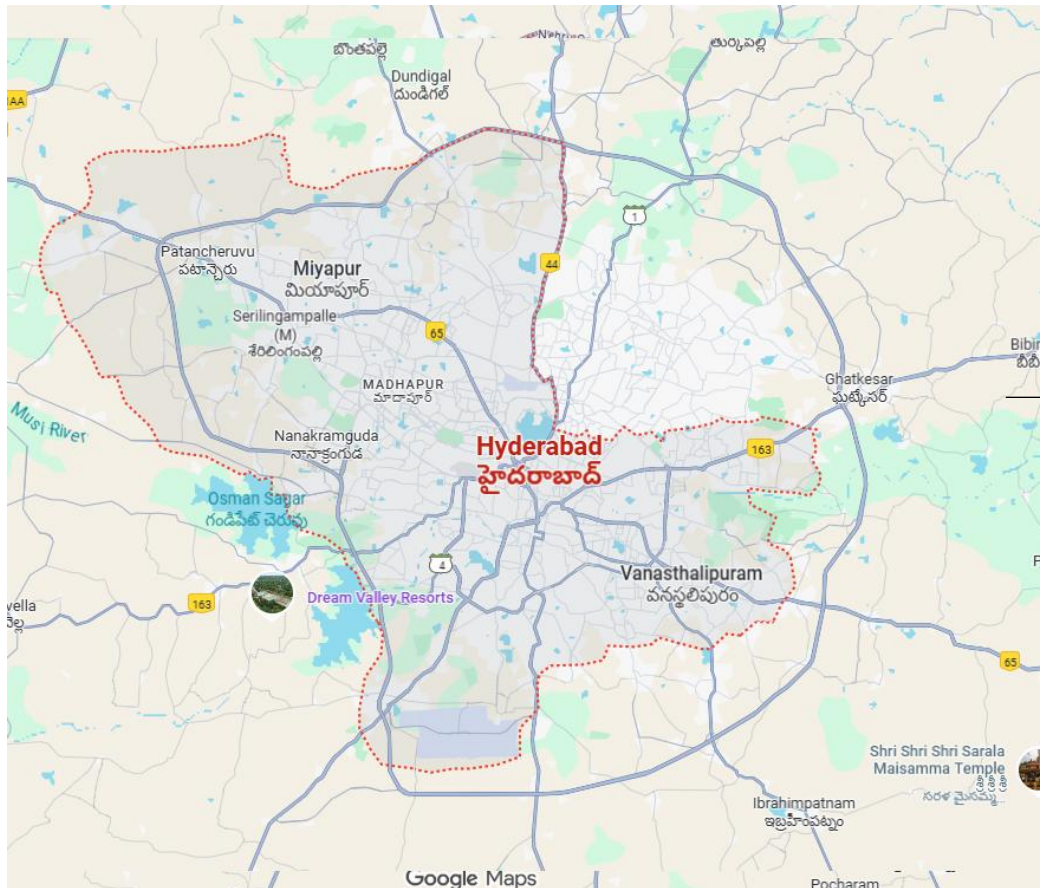






Congestion Measure Technique

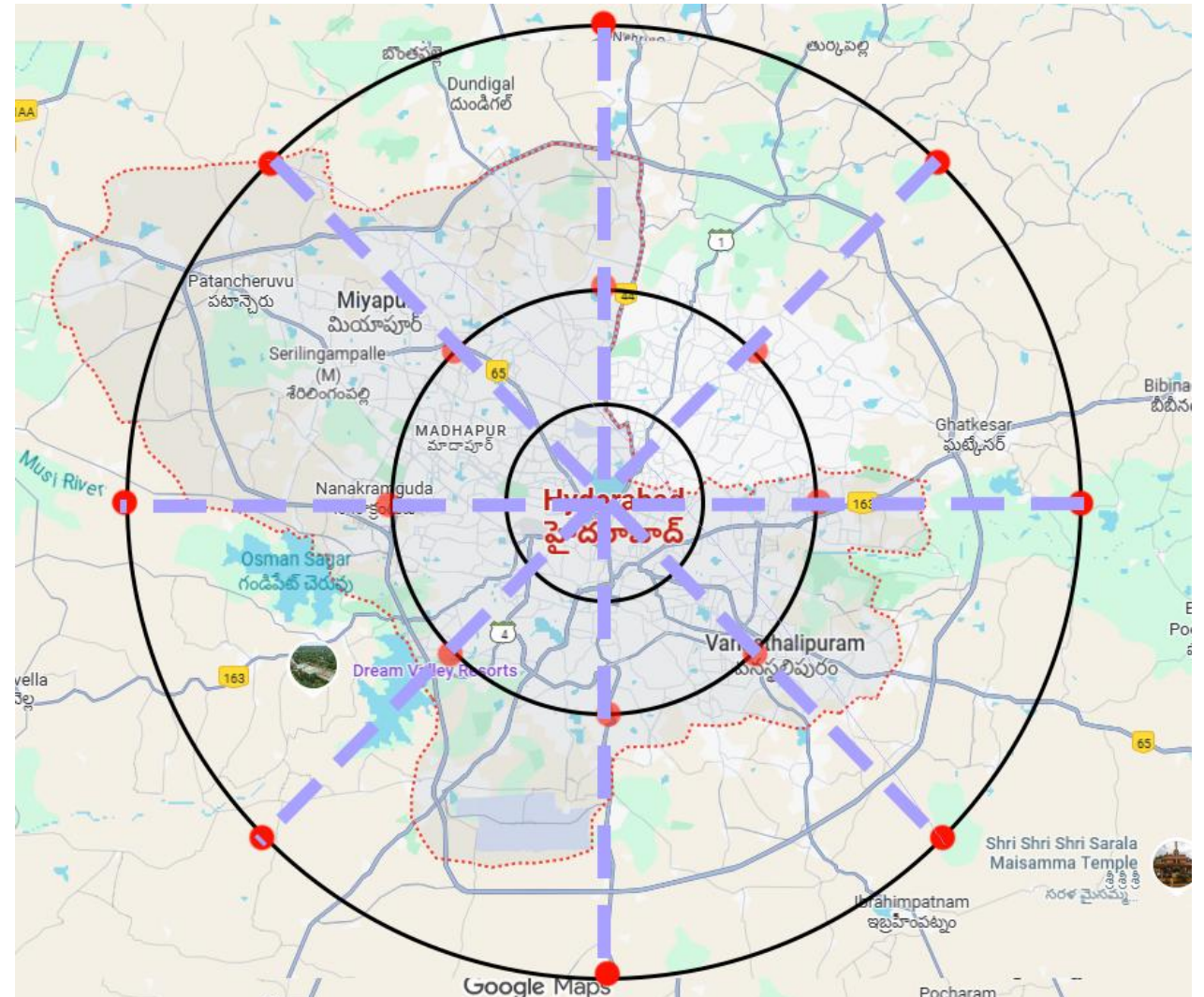
- Plot concentric circles centred on the city centre.
- 3 circles of radius 5, 10, 20 km.



- Plot points along the circles every 45 degrees.
- Measure travel time between all diametrically opposite points. (using Google Maps)
- Measure at peak time 9:00 AM, 6:00 PM
- Measure at non-peak time. 12:00 AM
- Measure Congestion Ratio.

$$\text{Congestion ratio} = \frac{\text{Travel Time}_{(\text{peak})}}{\text{Travel Time}_{(\text{non-peak})}}$$

- Take average across all points.
Calculate overall city congestion score.
- The same technique is applied for a neighbourhood to measure local congestion.



Formula Used to Calculate Score

Weighted Sum used with the following weights:

Category	Total Weight	% of Overall Score
Environment & Sustainability	25	25%
Infrastructure & Services	20	20%
Mobility & Accessibility	15	15%
Economy & Affordability	15	15%
Civic Engagement	10	10%
Total	85	100%

Category	Parameter	Weight	Notes
Environment & Sustainability (25)	Air Quality Index (AQI)	10	Negative weight (lower AQI = better)
	Water Quality Index (WQI)	7	Higher WQI = better
	Number of Parks	4	More parks = better livability
	Number of Playgrounds	4	Enhances recreational access
Infrastructure & Services (20)	Hospitals	6	Access to healthcare
	Schools	5	Education accessibility
	Supermarkets	3	Everyday convenience
	Stores	2	Local shopping access
	Libraries	4	Cultural & educational resource
Mobility & Accessibility (15)	City-level Congestion	9	Negative weight (less congestion = better)
	Local Congestion	6	Negative weight
Economy & Affordability (15)	Average Price per Sq.Ft	10	Negative weight (lower = better)
	HDI Rank	5	Negative weight (lower rank = better)
Civic Engagement (10)	Voter Turnout	10	Reflects citizen participation

Results:

Liveability of Hyderabad and Bangalore Visualized in an interactive map

Fig 1. Hyderabad

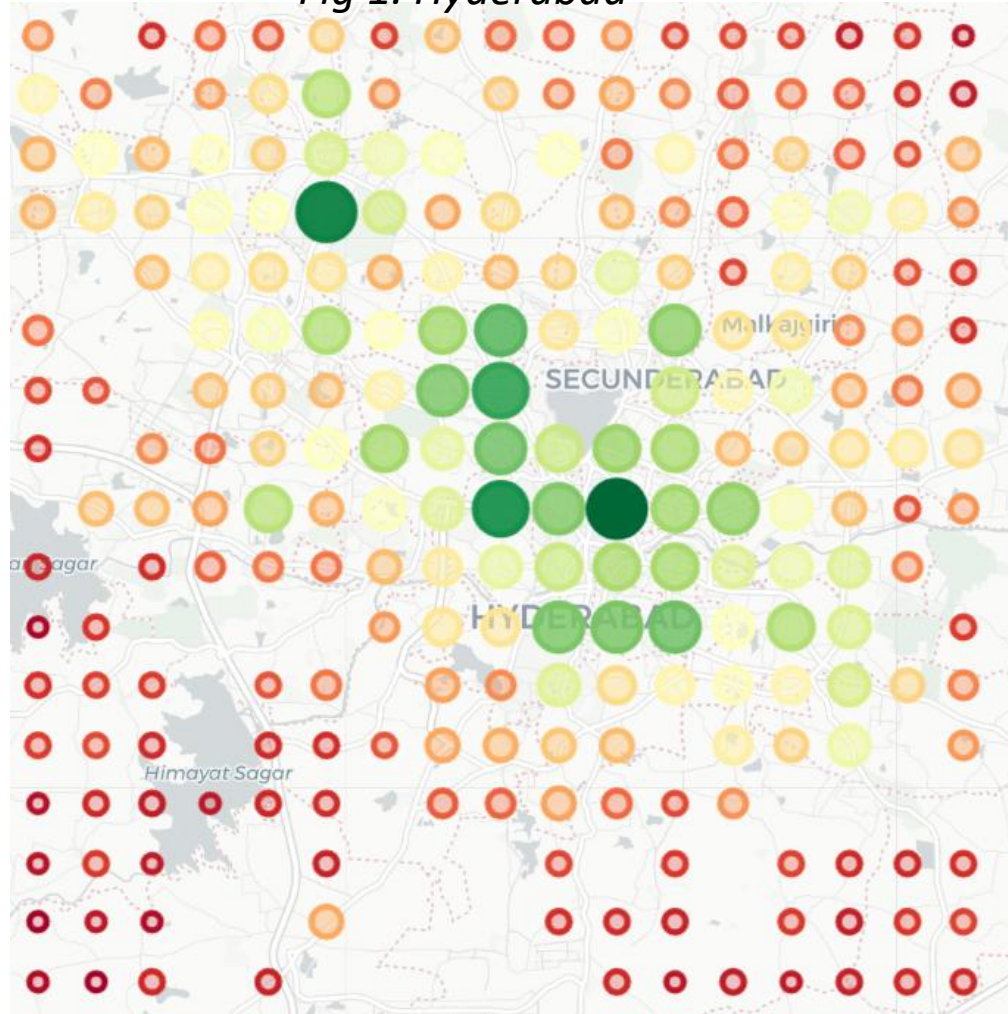


Fig 2. Bangalore

